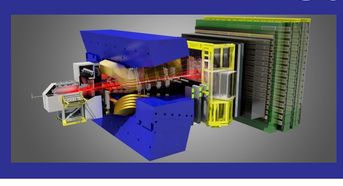
Linearity of Photodetectors for Calorimeter's upgrades at LHCb

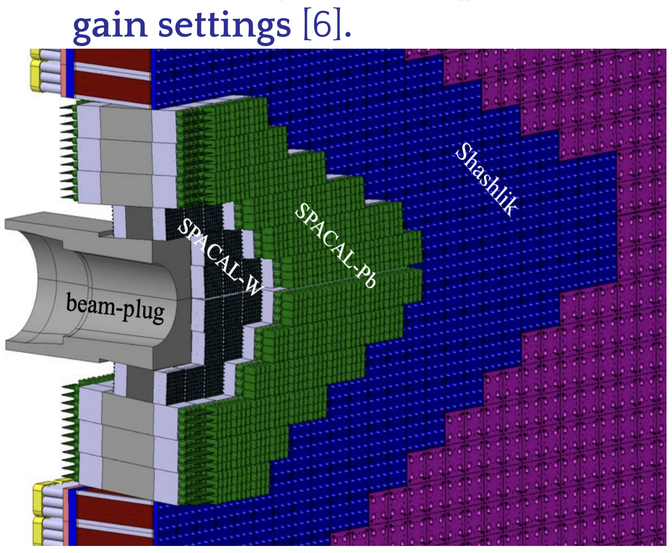
Othello D. Gomes[1], Elizabeth Kowalczyk[1], Emily Jiang[1], Phoebe M. Hamilton[1]

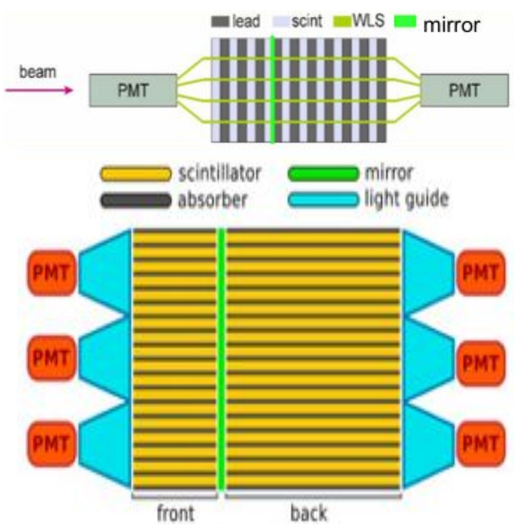
[1]*University of Maryland, Department of Physics*



Background

- The Large Hadron Collider beauty (LHCb) experiment's electromagnetic calorimeter (ECAL) provides measurements of the energies of gamma rays and electrons and aids with the identification of the latter.
- The present ECAL uses shashlik technology but the newer modules use SPACAL technology which will require faster & more radiant tolerant photodetectors [1][2][3].
- I present results which will determine ideal photodetector candidates by using a novel method to measure linearities (which is one of the primary key characteristics for mentioned candidate device [5]) of PMTs (photomultiplier tubes) at different voltage distribution schemes.
- PMTs must yield the specified linearity for the ECAL at lower gain settings [6].

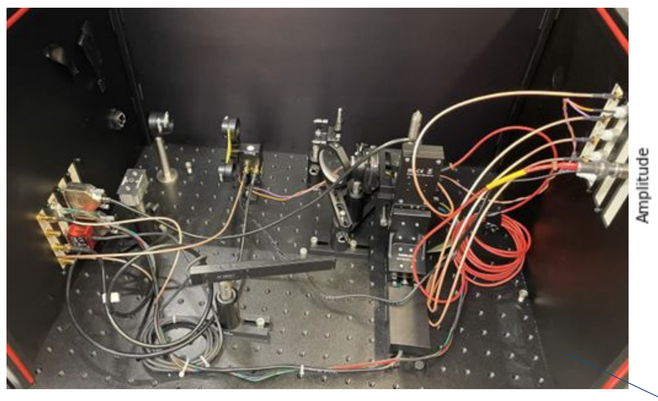


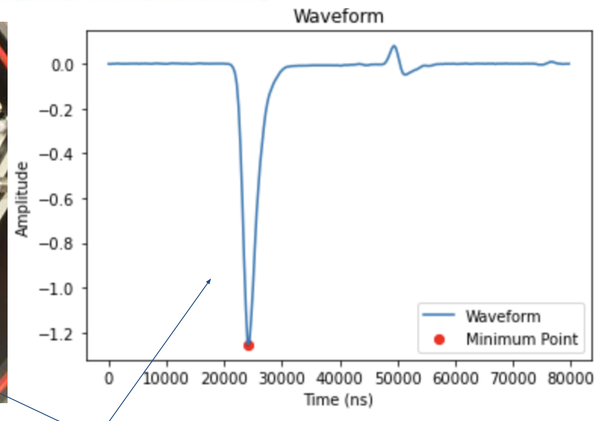


Setup

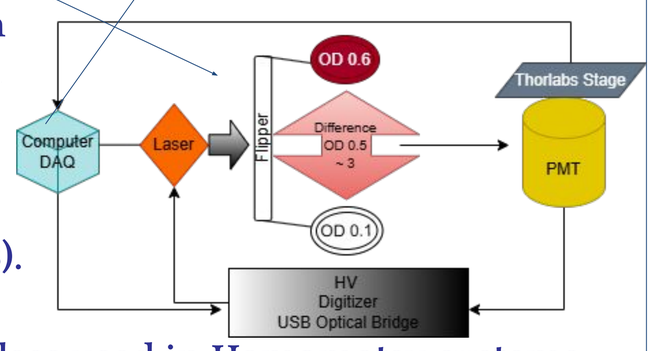
PMT	K	Dy1	Dy2	Dy3	Dy4	Dy5	Dy6	Dy7	Dy8	Dy9	Dy10	P
Uniform R9880U-20	1	1	1	1	1	1	1	1	1	1	0.5	
Tapered R9880U-20	2	1	1	1	1	1	1	2	3	4	3	
Zener R9880U-20	2	1	1	1	1	1	1	2	-75V	-100V	-75V	
K: Cathode, Dy: Dynode, P: Anode												

Table 1: Voltage distribution ratio for all variants of R9880U-20



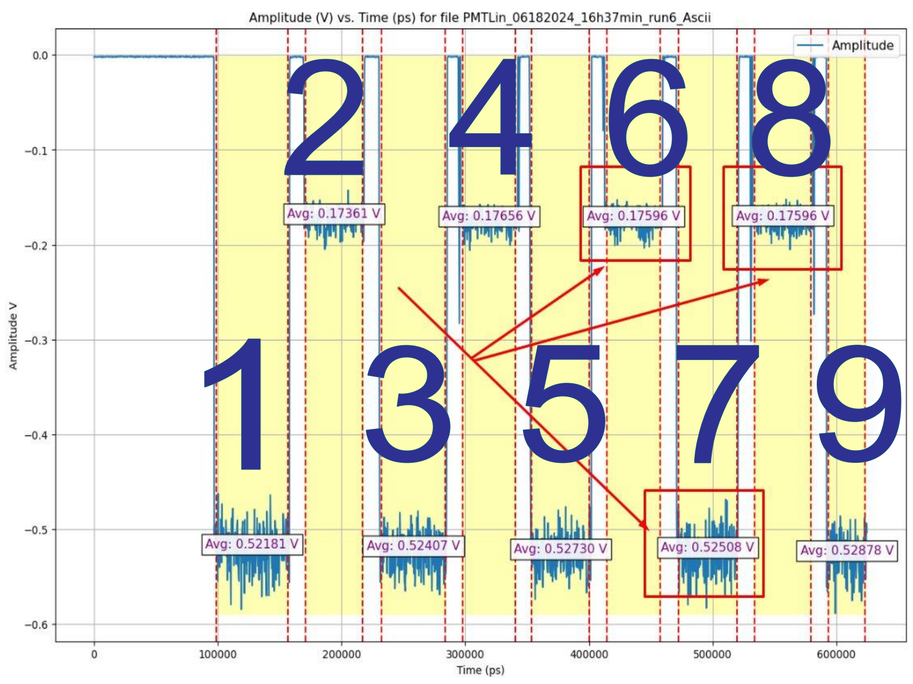


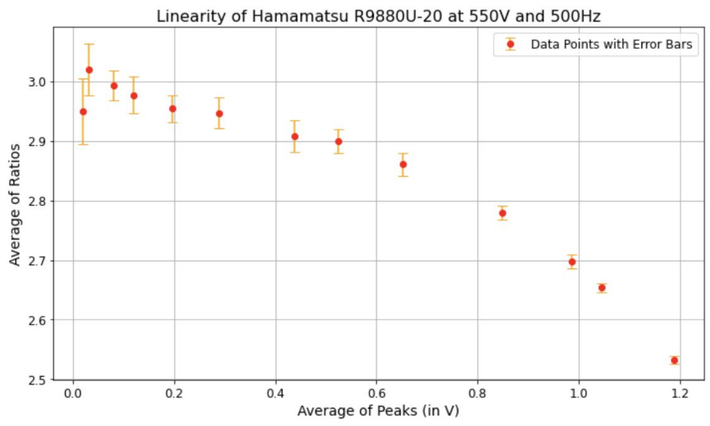
- A laser beam alternates between OD 0.1 and OD 0.6 filters using a flipper mount with a stopper to block the unfiltered beam, creating a brightness ratio difference of OD 0.5 (factor of ~3).

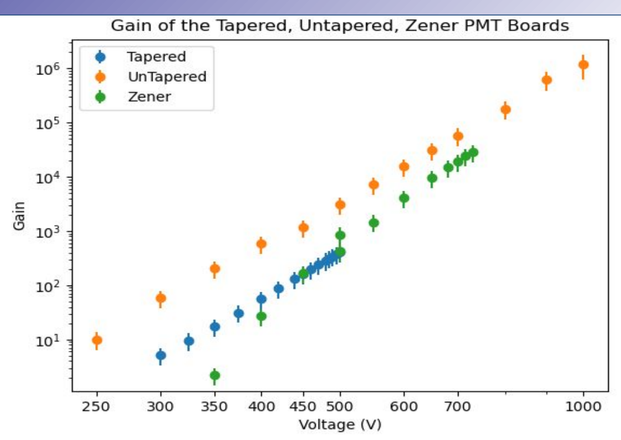


- This setup mimics dual LED pulses used in Hamamatsu systems [4] to test PMT linearity by analyzing the signal ratio from the waveforms of the two brightness levels.

Method

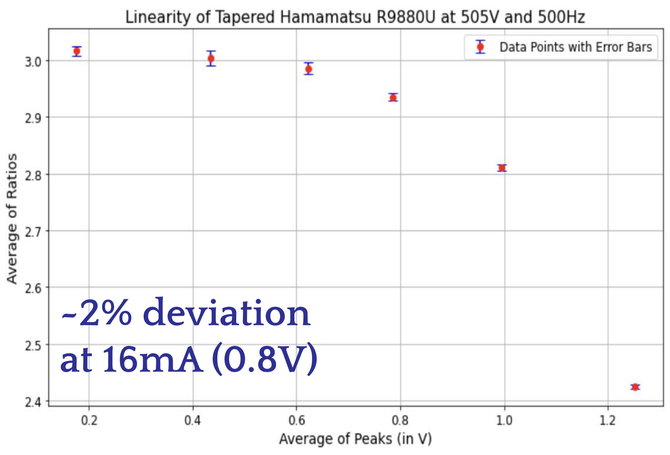


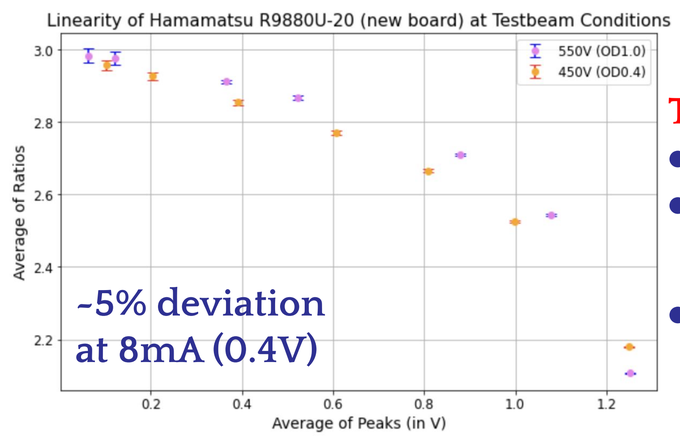


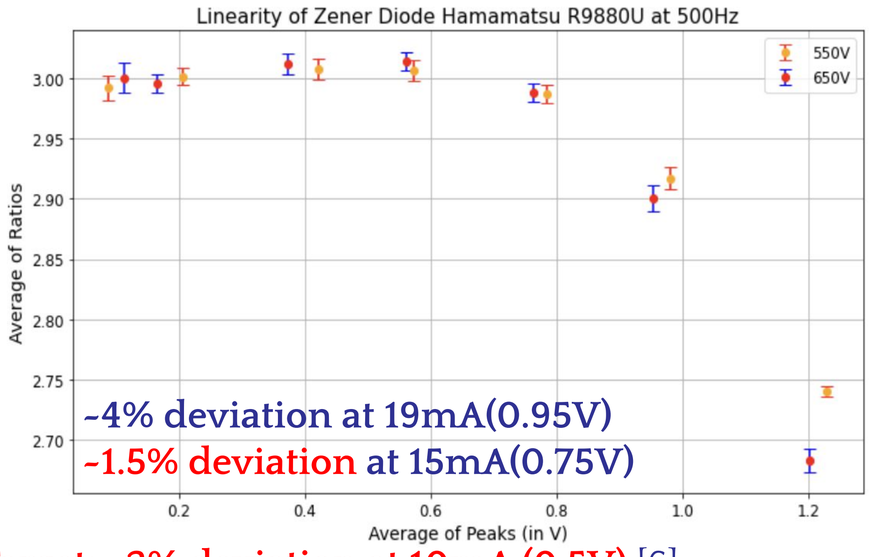


- A script is developed to isolate regions where the flipper is stationary by identifying continuous data segments below a threshold, avoiding periods when the flipper is moving.
- It computes the brightness ratios by averaging values in each stable region, dividing lower by higher intensity segments (e.g. 7 by 6 then 7 by 8), and then calculates the overall average ratio and uncertainty using the RMS of these values.
- The high voltage settings are chosen based on the gain of different voltage distribution schemes.

Results







Target: ~2% deviation at 10mA (0.5V) [6]

- Uniform divider shows the most nonlinear behavior.
- Tapered divider achieves linearity requirement at ~0.8V at a gain around 10^2 - 10^3
- Zener divider shows the best linearity out of all dividers, showing <2% non-linearity at around 0.75V with a similar gain setting to the tapered PMT.

Takeaways

- Zener diode provides the best linearity out of all variants by holding the voltages of the last few stages at maximum.
- We can achieve the specified linearity required for the ECAL's design at lower gain settings (1000-2000) with the Zener diode [6].
- R9880U survives the test in being a suitable candidate which Hamamatsu previously stated wasn't possible.
- Future work includes testing other PMTs like the R9800 and repeating the linearity test by the method of charge integration.

ACKNOWLEDGEMENTS: UMD LHCb Group, CERN LHCb Collaboration (in particular our ECAL collaborators, especially Matteo Salomoni, Eduardo Picatoste, Yuri Guz), Hamamatsu Photonics K.K.

References:

[1] **LHCb Collab.**, LHCb Upgrade Ib TDR, CERN-LHCC-2023-008 (2023); [2] **LHCb Collab.**, Upgrade II Framework TDR, CERN-LHCC-2022-011 (2022).
[3] **LHCb Collab.**, Upgrade II Scoping Doc., CERN-LHCC-2023-023 (2023); [4] **Hamamatsu**, PMT Handbook, 4th ed. (2007);
[5] **P. Horowitz** and W. Hill, The Art of Electronics, 3rd ed. (2015); [6] **A. Golutvin**, PMT Requirement List – PicoCal, CERN Indico (2024)